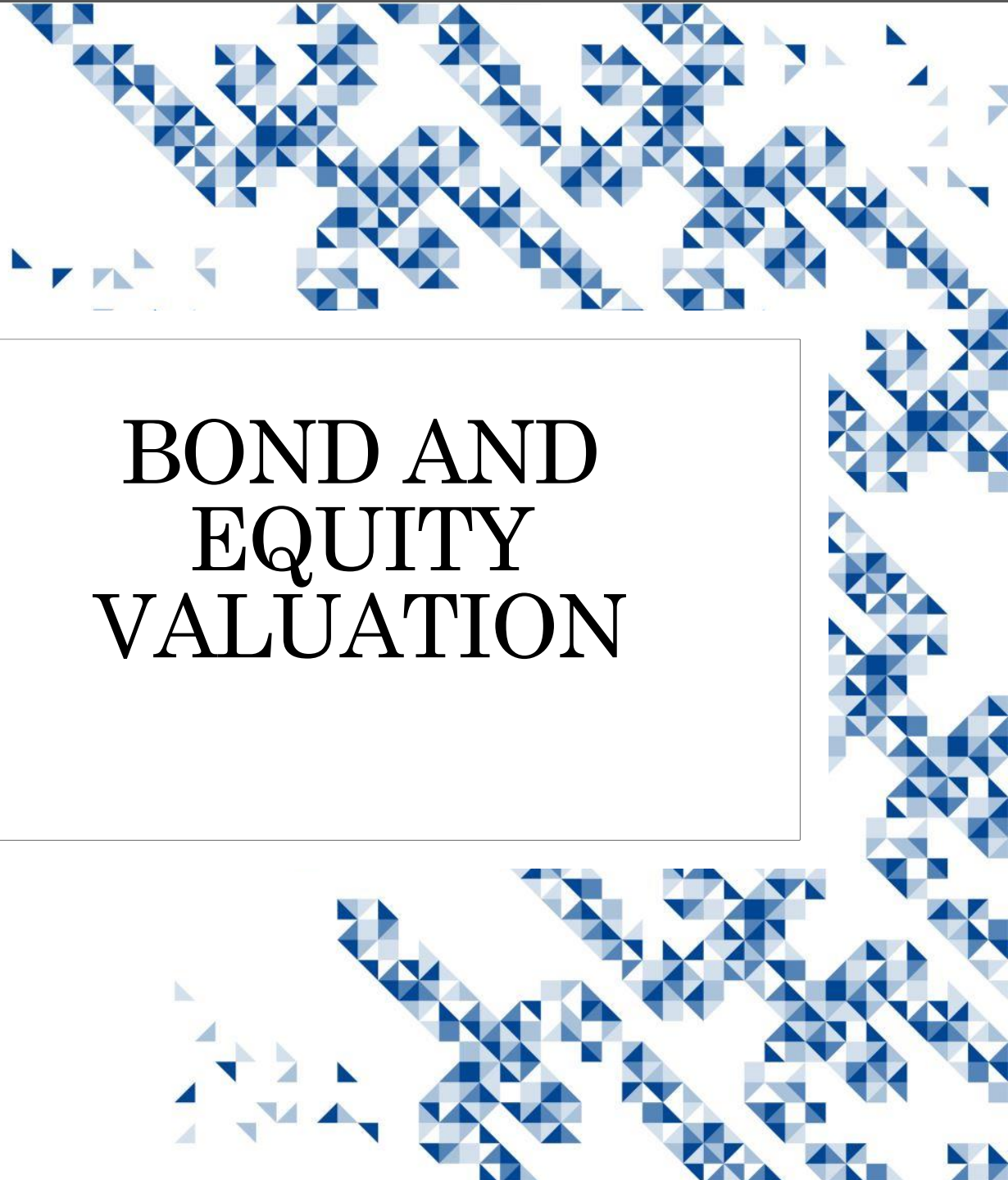


A decorative pattern of blue triangles and squares of varying shades, arranged in a complex, overlapping geometric design that frames the central text box.

BOND AND EQUITY VALUATION

Differences

- Debt instrument- Bonds or debentures
- Equity Instrument- Stocks or Shares
- Hybrid Instrument- Preferred Stocks or shares

Stocks vs Bonds

BASIS FOR COMPARISON	STOCKS	BONDS
Meaning	Stocks are the financial instrument that carries ownership interest, issued by the company in exchange for cash.	Bonds are the debt instrument issued by the companies to raise capital with a promise to pay back the money after some time along with interest.
Issued by	Companies	Government institutions, companies and financial institutions, etc.
What is it?	Equity instrument	Debt instrument
Return	Dividend	Interest
Is the return guaranteed?	No	Yes
Owners	Stockholders	Bondholders
Status of holders	Stockholders are the owners of the company.	Bondholders are the lenders to the company.
Risk	High	Comparatively low
Add on benefits	The holders get voting rights.	The holders get preference at the time of repayment.
Market	Centralized	Over the Counter (OTC)

Stock vs Preferred Stocks

ASPECT	COMMON STOCKS	PREFERRED STOCKS
DIVIDENDS	After the preferred stockholders are paid	Paid First
	Variable and may become higher	Pre-determined rates; unchanged
GROWTH	High potential but one that is connected to the company's growth	Don't grow as the company grows
LIQUIDATION	Paid last after all the creditors and preferred shareholders are paid	Given preference in terms of payment, just as with dividends
VOTING RIGHTS	Have them and can use them for deciding important issues	Don't have any voting rights
ARREARS	Missed dividends don't accrue, so they can be paid the next year	Accrue of the arrears added to the next year's payment
CERTAINTY	No profits mean no dividends	Paid even if the company incurs losses

Cash flows from instruments

- Common Stocks- Dividends and Stock Price (if sold)
- Preferred Stock- Preference Dividend and Amount paid at maturity
- Bond/Debenture- Interest received and maturity payment for principal

Bond Valuation

- A bond represents a contract under which a borrower promises to pay interest and principal on specific dates to the holders of the bond
- **Par Value:** This is the value stated on the face of the bond. It represents the amount the firm borrows and promises to repay at the time of maturity.
- **Coupon Rate and Interest:** A bond carries a specific interest rate which is called 'the coupon rate'. The interest payable to the bond holder is simply: **par value of the bond x coupon rate**.
- **Maturity Period**

Bond types

- Coupon Bond
- Zero Coupon Bond
- Perpetual Bonds

NSE Traded Debt/Fixed-Income Securities



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SYMBOL	SERIES	BOND TYPE	COUPON RATE	FACE VALUE	LTP	%CHNG	VOLUME (Shares)	VALUE (₹ Lakhs)	CREDIT RATING	MATURITY DATE
RECLTD	NI	Tax Free	8.12	1,000	1,238.25	-1.53	9,400	116.38	AAA/Stable	27-Mar-2027
NHAI	N2	Tax Free	8.30	1,000	1,245.00	0.40	9,162	113.93	AAA/Stable;AAA;AAA(ind)	25-Jan-2027
NTPC	N7	Regular	8.49	13	14.56	0.00	7,44,700	106.86	-	25-Mar-2025
SBIN	N5	Regular	9.95	10,000	10,995.00	-0.08	581	63.91	AAA;AAA/Stable	16-Mar-2026
NHAI	N1	Tax Free	8.20	1,000	1,065.90	0.15	4,850	51.65	AAA/Stable;AAA;AAA(ind)	25-Jan-2022
IRFC	NK	Tax Free	7.50	1,000	1,321.75	-0.06	2,870	37.95	AAA/Stable	21-Dec-2035
HUDCO	N2	Tax Free	8.20	1,000	1,308.10	0.25	2,452	32.02	AA+(ind);AA+	05-Mar-2027
TCFSL	ND	Regular	8.90	1,000	1,096.00	-0.84	2,168	23.82	CRISIL AAA / CARE AAA	27-Sep-2023
NTPC	ND	Tax Free	7.62	1,000	1,325.00	0.00	1,555	20.60	AAA	05-Oct-2035
NHAI	NA	Tax Free	7.60	1,000	1,294.94	0.22	1,585	20.50	AAA/Stable	11-Jan-2031
HUDCO	NE	Tax Free	9.01	1,000	1,422.00	-0.36	1,385	19.73	AA+	13-Jan-2034
NHAI	N6	Regular	8.75	1,000	1,369.00	0.81	1,225	16.70	AAA/STABLE;AAA;AAA	15-Feb-2029
IFCI	NH	Regular	9.90	1,000	1,050.00	0.48	1,411	14.74	BWR AA-/Stable, ICRA A/Stable	01-Dec-2024

NHAI Example



Search by company name, symbol or keyword

 **NSE** Debt Market

Debt Market is Closed

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N2

25-Jan-2027

Clear

SERIES	ISIN	ISSUE DESCRIPTION	MATURITY DATE	OPEN PRICE	CLOSE PRICE	PREV. CLOSE	LAST PRICE	CHNG	%CHNG	VOLUME (Shares)	VALUE (₹ Lakhs)
N2	INE906B07CB9	BOND 8.30% PA Tax Free S2	25-Jan-2027	1,241.99	1,244.95	1,240.10	1,245.00	4.90	0.40	9,162	113.93

Trade Information

Intraday Chart

Securities Information

BOND TYPE	ISSUE DESCRIPTION	ISSUE DATE	MATURITY DATE	COUPON RATE	NEXT IP DATE	FACE VALUE	CREDIT RATING
Tax Free	BOND 8.30% PA Tax Free S2	25-Jan-2012	25-Jan-2027	8.30	01-Oct-2016	1,000	AAA/Stable;AAA;AAA(ind)

Price Information

1,245.00

4.90 (0.40%)

PREV. CLOSE	OPEN	CLOSE*	VWAP	LOWER BAND	UPPER BAND	PRICE BAND	ADJUSTED PRICE *
1,240.10	1,241.99	1,244.95	1,243.55	992.08	1,488.12	NO BAND	-

INTRADAYHIGH LOW

1,245.00

52 WEEKHIGH LOW

1,245.00

Valuing a bond: Annual Coupon

$$P = \sum_{t=1}^n \frac{C}{(1+r)^t} + \frac{M}{(1+r)^n}$$

where P is the value (in rupees), n is the number of years to maturity, C is the annual coupon payment (in rupees), r is the periodic required return, M is the maturity value, and t is the time when the payment is received.

$$P = C \times \text{PVIFA}_{r,n} + M \times \text{PVIF}_{r,n}$$

Example

- Compute the price of a bond, consider a 10-year, 12% coupon bond with a par value of Rs. 1,000. Let us assume that the required yield on this bond is 13%.

t=0	120	120	120	120	120 +1000
	120	120	120	120	120

Solution

	CP=12%	13% Rate	
Year	Cash Flow	PVIF	PV CashFlow
1	120	0.884956	106.195
2	120	0.783147	93.978
3	120	0.69305	83.166
4	120	0.613319	73.598
5	120	0.54276	65.131
6	120	0.480319	57.638
7	120	0.425061	51.007
8	120	0.37616	45.139
9	120	0.332885	39.946
10	1120	0.294588	329.939
			945.738

	CP=12%	10% Rate	
Year	Cash Flow	PVIF	PV CashFlow
1	120	0.909091	109.091
2	120	0.826446	99.174
3	120	0.751315	90.158
4	120	0.683013	81.962
5	120	0.620921	74.511
6	120	0.564474	67.737
7	120	0.513158	61.579
8	120	0.466507	55.981
9	120	0.424098	50.892
10	1120	0.385543	431.808
			1122.891

	CP=12%	12% Rate	
Year	Cash Flow	PVIF	PV CashFlow
1	120	0.892857	107.143
2	120	0.797194	95.663
3	120	0.71178	85.414
4	120	0.635518	76.262
5	120	0.567427	68.091
6	120	0.506631	60.796
7	120	0.452349	54.282
8	120	0.403883	48.466
9	120	0.36061	43.273
10	1120	0.321973	360.610
			1000.000

$$P = 120 \times PVIFA_{13\%,10\text{yrs}} + 1,000 \times PVIF_{13\%,10\text{yrs}}$$

Semi-annual coupon

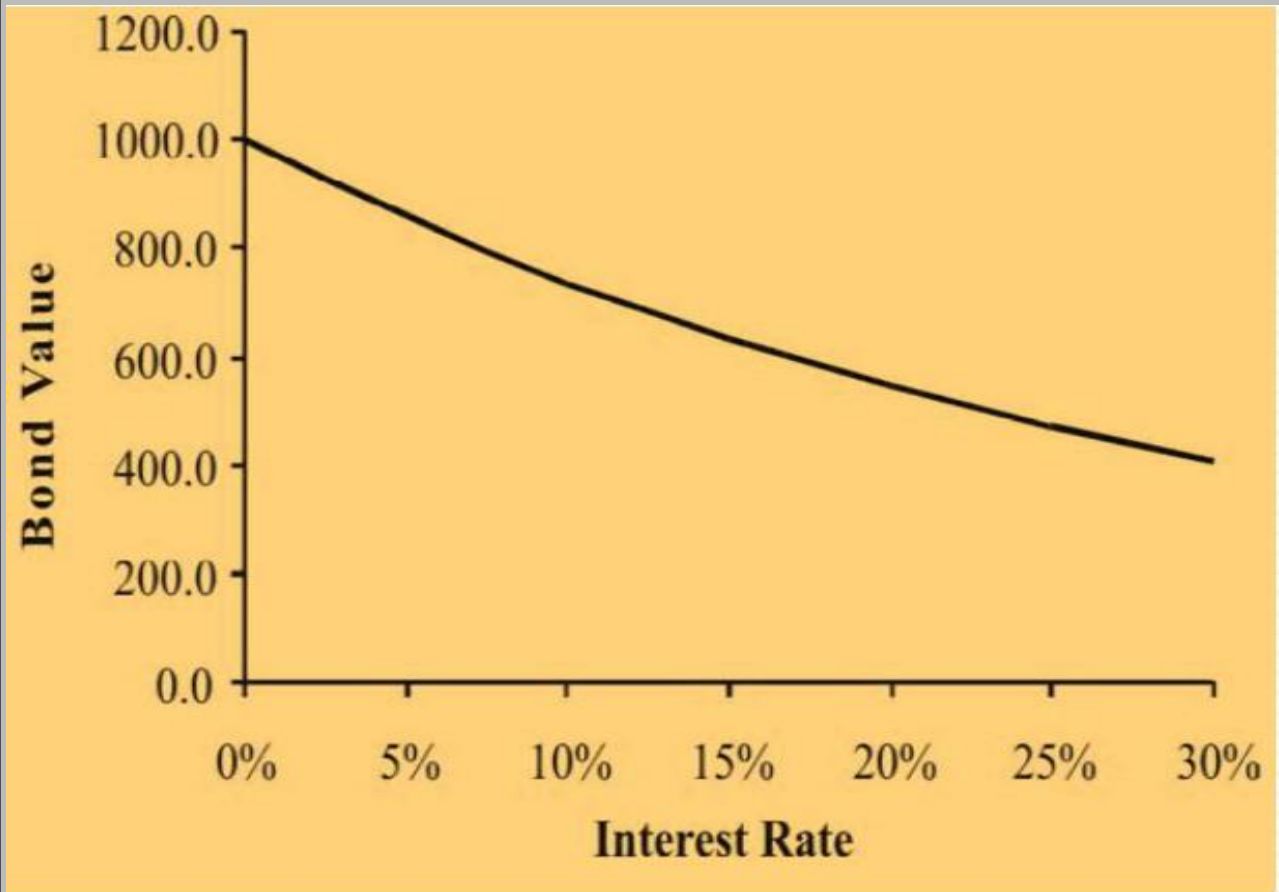
$$\begin{aligned} P &= \sum_{t=1}^{2n} \frac{C/2}{(1 + r/2)^t} + \frac{M}{(1 + r/2)^{2n}} \\ &= C/2 (\text{PVIFA}_{r/2, 2n}) + M(\text{PVIF}_{r/2, 2n}) \end{aligned}$$

Example

- Compute the price of a bond, consider a 10-year, 12% coupon bond paid semi-annually with a par value of Rs. 1,000. Let us assume that the required yield on this bond is 13%.

Interest	PVFA	60*PVFA	PVIF	PV_M	PV of Bond
13.00%	11.019	661.110	0.284	283.797	944.907
12.00%	11.470	688.195	0.312	311.805	1000.000
10.00%	12.462	747.733	0.377	376.889	1124.622

Bond Value and Interest Rate

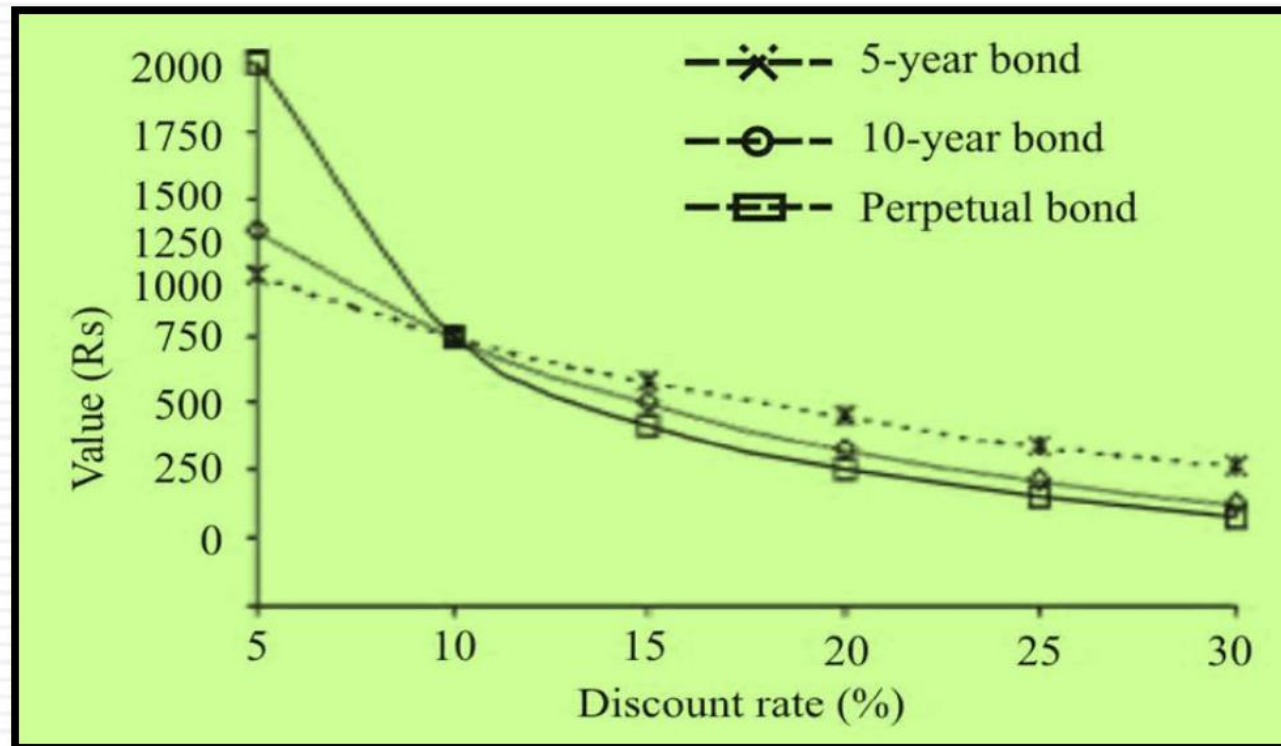


The value of the bond declines as the market interest rate (discount rate) increases.

The value of a 10-year, 12 per cent Rs 1,000 bond for the market interest rates ranging from 0 per cent to 30 per cent is shown in the figure.

Bond value, maturity period and interest rate

Bond Maturity and Interest Rate Risk



Yield to Maturity (YTM)

- The return or yield the bond holder will earn on the bond if he or she buys it at its current market price, receives all coupon and principal payments as promised, and holds the bond until maturity.
- YTM for bond coupon paid semi-annually-

$$\begin{aligned} V_b &= \frac{INT/2}{(1 + ytm/2)^1} + \frac{INT/2}{(1 + ytm/2)^2} + \dots + \frac{INT/2}{(1 + ytm/2)^{2T}} + \frac{M}{(1 + ytm/2)^{2T}} \\ &= \frac{INT}{2} \left[\frac{1 - \frac{1}{(1 + ytm/2)^{2T}}}{ym/2} \right] + 1[1 + (ym/2)]^{2T} \end{aligned}$$

Example

$$YTM = \frac{C + \frac{F - V}{n}}{\frac{F + V}{2}}$$

You are considering the purchase of a \$1,000 face value bond that pays 11 percent coupon interest per year, paid semiannually (i.e., \$55 (= \$1,000(.11)/2) per semiannual period). The bond matures in 15 years and has a face value of \$1,000. If the current market price of the bond is \$931.176, the yield to maturity, or $E(r_b)$ is calculated as follows:

$$931.176 = \frac{1,000(.11)}{2} \left[\frac{1 - \frac{1}{(1 + ytm/2)^{2(15)}}}{ym/2} \right] + 1,000/(1 + ytm/2)^{2(15)}$$

$$YTM_{\text{approximate}} = \frac{C + (FP - MP)/n}{(FP + MP)/2}$$

YTM → Yield to Maturity

C → coupon payment or annual fixed interest

FP → face value of the bond

MP → market price of bond

n → number of years to maturity

Bond Rating

		Investment-Quality Bond Ratings				Low-Quality, Speculative, and/or “Junk” Bond Ratings					
		High Grade		Medium Grade		Low Grade		Very Low Grade			
Standard & Poor’s		AAA	AA	A	BBB	BB	B	CCC	CC	C	D
Moody’s		Aaa	Aa	A	Baa	Ba	B	Caa	Ca	C	
Moody’s	S&P										
Aaa	AAA	Debt rated Aaa and AAA has the highest rating. Capacity to pay interest and principal is extremely strong.									
Aa	AA	Debt rated Aa and AA has a very strong capacity to pay interest and repay principal. Together with the highest rating, this group comprises the high-grade bond class.									
A	A	Debt rated A has a strong capacity to pay interest and repay principal, although it is somewhat more susceptible to the adverse effects of changes in circumstances and economic conditions than debt in high-rated categories.									
Baa	BBB	Debt rated Baa and BBB is regarded as having an adequate capacity to pay interest and repay principal. Whereas it normally exhibits adequate protection parameters, adverse economic conditions or changing circumstances are more likely to lead to a weakened capacity to pay interest and repay principal for debt in this category than in higher-rated categories. These bonds are medium-grade obligations.									
Ba; B Caa Ca C	BB; B CCC CC C	Debt rated in these categories is regarded, on balance, as predominantly speculative with respect to capacity to pay interest and repay principal in accordance with the terms of the obligation. BB and Ba indicate the lowest degree of speculation, and Ca, CC, and C the highest degree of speculation. Although such debt is likely to have some quality and protective characteristics, these are outweighed by large uncertainties or major risk exposures to adverse conditions. Issues rated C by Moody’s are typically in default.									
	D	Debt rated D is in default, and payment of interest and/or repayment of principal is in arrears.									

Problem

- You are considering the purchase of a \$1,000 face value bond that pays 10 percent coupon interest per year, with the coupon paid semiannually. The bond matures in 12 years and the required rate of return (r_b) on this bond is 8 percent. the value of the bond is ???

Problem

- You are considering the purchase of a \$1,000 face value bond that pays 10 percent coupon interest per year, with the coupon paid semiannually. The bond matures in 12 years and the required rate of return (r_b) on this bond is **8 percent**

$$\begin{aligned} V_b &= \frac{1,000(.1)}{2} \left[\frac{1 - \frac{1}{[1 + (.08/2)]^{2(12)}}}{.08/2} \right] + 1,000/[1 + (.08/2)]^{2(12)} \\ &= 50(15.24696) + 1,000(.39012) = \$1,152.47 \end{aligned}$$

- You are considering the purchase of a \$1,000 face value bond that pays 10 percent coupon interest per year, with the coupon paid semiannually. The bond matures in 12 years and the required rate of return (*rb*) on this bond is **10 percent**

$$V_b = \frac{1,000(.1)}{2} \left[\frac{1 - \frac{1}{[1 + (.10/2)]^{2(12)}}}{.10/2} \right] + 1,000/[1 + (.10/2)]^{2(12)}$$

$$= 50(13.79864) + 1,000(.31007) = \$1,000.00$$

- If required rate of return

$$V_b = \frac{1,000(.1)}{2} \left[\frac{1 - \frac{1}{[1 + (.12/2)]^{2(12)}}}{.12/2} \right] + 1,000/[1 + (.12/2)]^{2(12)}$$

$$= 50(12.55036) + 1,000(.24698) = \$874.50$$

Pure Discount/Zero Coupon Bond

Pure discount bond do not carry an explicit rate of interest.

It provides for the payment of a lump sum amount at a future date in exchange for the current price of the bond.

The difference between the face value of the bond and its purchase price gives the return or *YTM* to the investor.

Example: A company may issue a pure discount bond of Rs 1,000 face value for Rs 520 today for a period of five years. The rate of interest can be calculated as follows:

$$520 = \frac{1,000}{(1 + YTM)^5}$$

$$(1 + YTM)^5 = \frac{1,000}{520} = 1.9231$$

$$i = 1.9231^{1/5} - 1 = 0.14 \text{ or } 14\%$$

Perpetual Bond

→ bond with no maturity date
↳ infinite series

A perpetual bond's yield-to-maturity:

$$B_0 = \sum_{t=1}^{n=\infty} \frac{\text{INT}}{(1+k_d)^t} = \frac{\text{INT}}{k_d}$$

- For example, if a perpetual bond pays \$100 per year in perpetuity and the discount rate is assumed to be 4%, the present value would be:
- Present value = \$100 / 0.04 = \$2500
- Example: Reliance Ind. Ltd- issued 100 years Yankee Bond in 1997 maturing 2097 with convertibility option

Bond Prices and Yields

- Prices and yields have an inverse relationship
- The bond price curve is convex
- The longer the maturity → the more sensitive the bond's price to changes in market interest rates

The Inverse Relationship Between Bond Prices and Yields

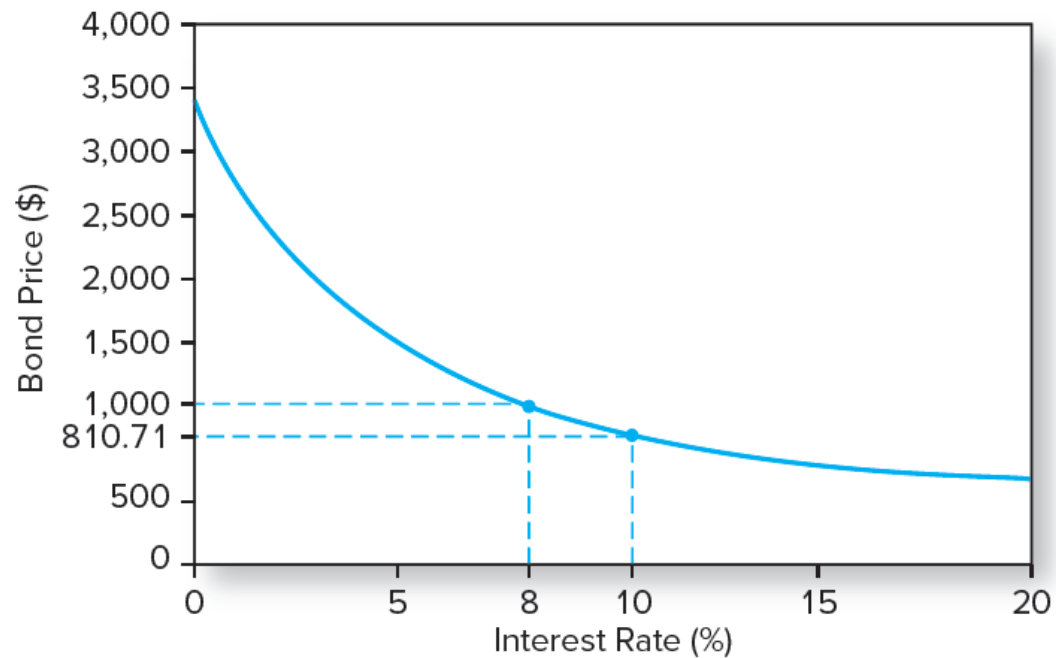


Figure 14.3 The inverse relationship between bond prices and yields. Price of an 8% coupon bond with 30-year maturity making semiannual payments

Bond Prices at Different Interest Rates

Table 14.2

Time to Maturity	Bond Price at Given Market Interest Rate				
	2%	4%	6%	8%	10%
1 year	1,059.11	1,038.83	1,019.13	1,000.00	981.41
10 years	1,541.37	1,327.03	1,148.77	1,000.00	875.35
20 years	1,985.04	1,547.11	1,231.15	1,000.00	828.41
30 years	2,348.65	1,695.22	1,276.76	1,000.00	810.71

Bond prices at different interest rates (8% coupon bond, coupons paid semiannually)

Yield to Maturity (YTM)

- The return or yield the bond holder will earn on the bond if he or she buys it at its current market price, receives all coupon and principal payments as promised, and holds the bond until maturity.
- YTM for bond coupon paid semi-annually-

$$\begin{aligned} V_b &= \frac{INT/2}{(1 + ytm/2)^1} + \frac{INT/2}{(1 + ytm/2)^2} + \dots + \frac{INT/2}{(1 + ytm/2)^{2T}} + \frac{M}{(1 + ytm/2)^{2T}} \\ &= \frac{INT}{2} \left[\frac{1 - \frac{1}{(1 + ytm/2)^{2T}}}{ym/2} \right] + 1[1 + (ym/2)]^{2T} \end{aligned}$$

Example

You are considering the purchase of a \$1,000 face value bond that pays 11 percent coupon interest per year, paid semiannually (i.e., \$55 (= \$1,000(.11)/2) per semiannual period). The bond matures in 15 years and has a face value of \$1,000. If the current market price of the bond is \$931.176, the yield to maturity, or $E(r_b)$ is calculated as follows:

$$931.176 = \frac{1,000(.11)}{2} \left[\frac{1 - \frac{1}{(1 + ytm/2)^{2(15)}}}{ym/2} \right] + 1,000/(1 + ytm/2)^{2(15)}$$

The yield to maturity is the nominal return on the bond. Its effective annual return is calculated as

$$EAR = (1 + ytm/2)^2 - 1 = (1 + .12/2)^2 - 1 = 12.36\%$$

Example

You are considering the purchase of a \$1,000 face value bond that pays 11 percent coupon interest per year, paid semiannually (i.e., \$55 (= \$1,000(.11)/2) per semiannual period). The bond matures in 15 years and has a face value of \$1,000. If the current market price of the bond is \$931.176, the yield to maturity, or $E(r_b)$ is calculated as follows:

$$931.176 = \frac{1,000(.11)}{2} \left[\frac{1 - \frac{1}{(1 + ytm/2)^{2(15)}}}{ym/2} \right] + 1,000/(1 + ytm/2)^{2(15)}$$

The yield to maturity is the nominal return on the bond. Its effective annual return is calculated as

$$EAR = (1 + ytm/2)^2 - 1 = (1 + .12/2)^2 - 1 = 12.36\%$$

$$YTM_{\text{approximate}} = \frac{C + (FP - MP)/n}{(FP + MP)/2}$$

YTM $\longrightarrow$ Yield to Maturity

C $\longrightarrow$ coupon payment or annual fixed interest

FP $\longrightarrow$ face value of the bond

MP $\longrightarrow$ market price of bond

n $\longrightarrow$ number of years to maturity

Yield to Maturity Example

- Suppose an 8% coupon, 30 year bond is selling for \$1276.76. What is its average rate of return?

$$\$1276.76 = \sum_{t=1}^{60} \frac{\$40}{(1+r)^t} + \frac{1000}{(1+r)^{60}}$$

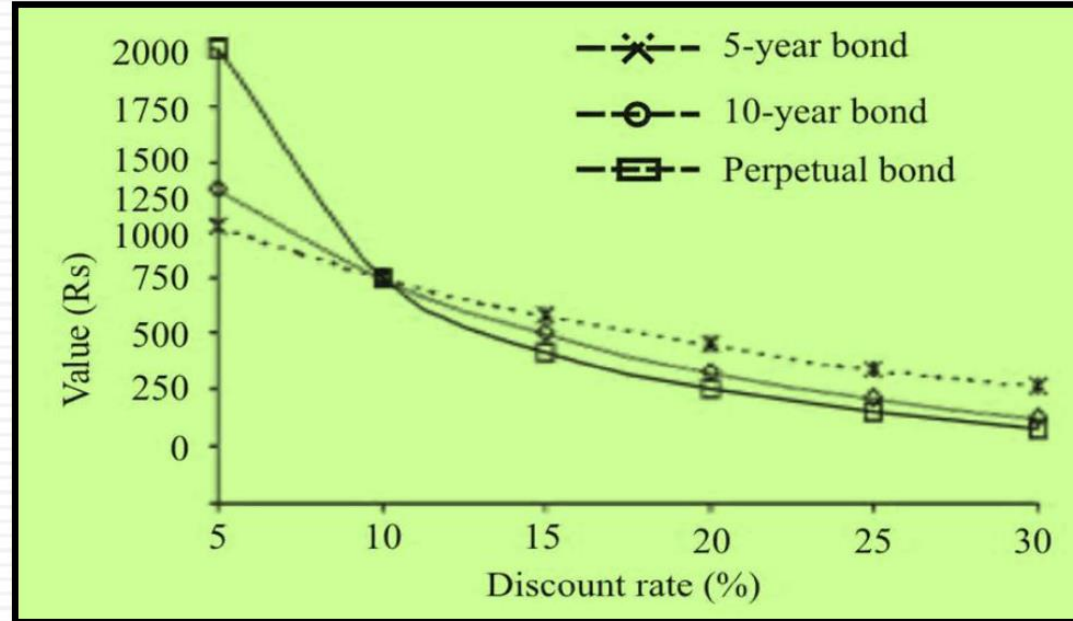
- $r = 3\%$ per half year
- Bond equivalent yield = 6%
- $\text{EAR} = ((1.03)^2) - 1 = 6.09\%$

Bond Yields: YTM vs. Current Yield

- Yield to Maturity
 - Bond's internal rate of return
 - The interest rate $\rightarrow$ PV of a bond's payments equal to its price
 - Assumes that all bond coupons can be reinvested at the YTM
- Current Yield:
 - Bond's annual coupon payment divided by the bond price
- Premium Bonds: *Coupon rate* $>$ *Current yield* $>$ *YTM*
- Discount Bonds: *Coupon rate* $<$ *Current yield* $<$ *YTM*

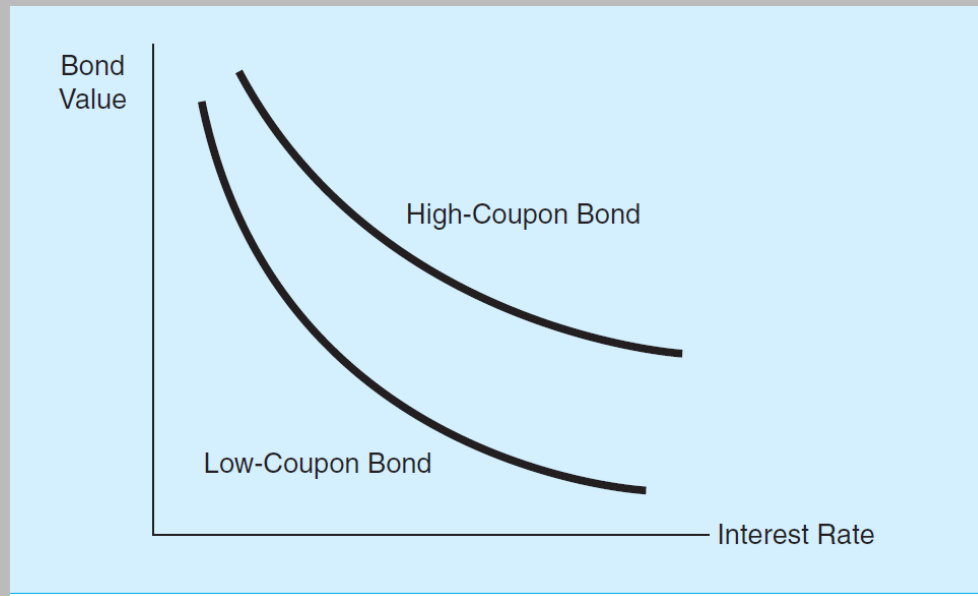
Bond value, maturity period and interest rate

Bond Maturity and Interest Rate Risk



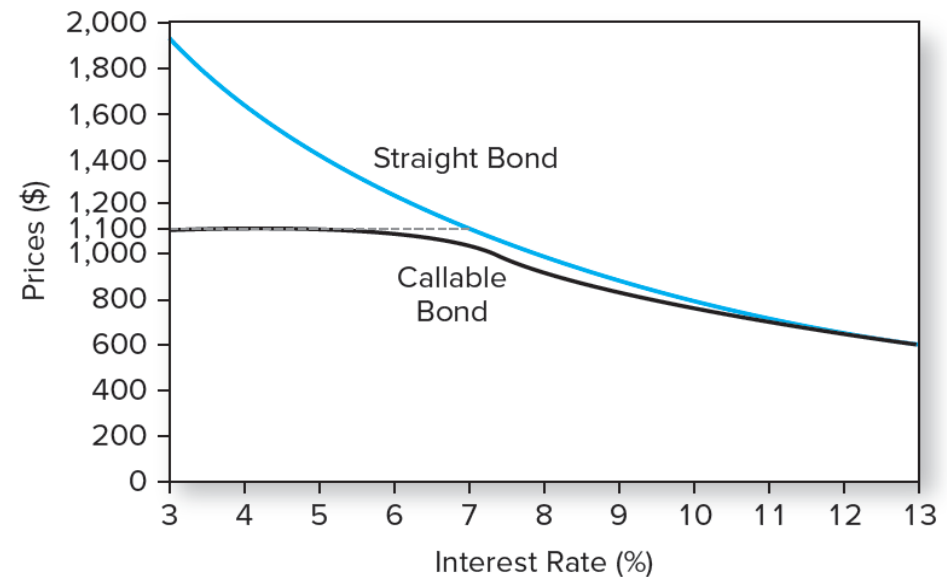
Coupon Rate and Security Price Sensitivity to Changes in Interest Rates

The Impact of a Bond's Coupon Rate on Its Interest Rate Sensitivity



Bond Yields: Yield to Call

- Low Interest Rates: The price of the callable bond is flat since the risk of repurchase or call is high
- High Interest Rates: The price of the callable bond converges to that of a normal bond since the risk of call is negligible



Callable at 110% of par value

Figure 14.4 Bond prices: Callable and straight debt (coupon = 8%; maturity = 30 years; semiannual payments)

Yield to Call (YTC)

For calculating the **yield to call**, the call period would be different from the maturity period and the call (or redemption) value could be different from the maturity value.

Example: Suppose the 10% 10-year Rs 1,000 bond is redeemable (callable) in 5 years at a call price of Rs 1,050. The bond is currently selling for Rs 950. The bond's yield to call is 12.7%.

$$950 = \sum_{t=1}^5 \frac{100}{(1 + \text{YTC})^t} + \frac{1,050}{(1 + \text{YTC})^5}$$

Measures of Bond

- Intrinsic Value
- Yield to Maturity (YTM)
- Yield to Call (YTC)
- Current Yield

→ measure of interest rate sensitivity of a bond

Duration of Bond

- The effect of maturity and coupon rates on the sensitivity of bond prices to changes in interest rates is complex to deal and understand
- Duration provides a simple measure that allows for a straightforward calculation of a bond's interest rate sensitivity.
- Duration is the weighted - average time to maturity on a financial security using the relative present values of the cash flows as weights. On a time value of money basis, duration measures the weighted average of when cash flows are received on a security.

Duration (Macaulay's Duration) Calculation

$$D = \frac{\sum_{t=1}^N \frac{CF_t \times t}{(1+r)^t}}{\sum_{t=1}^N \frac{CF_t}{(1+r)^t}} = \frac{\sum_{t=1}^N PV_t \times t}{\sum_{t=1}^N PV_t}$$

where

D = Duration measured in years

t = 1 to T , the period in which a cash flow is received

N = Number of periods to maturity

CF_t = Cash flow received on the security at end of period t

r = Current required rate of return (r) or yield to maturity (ym) on the investment

PV_t = Present value of the cash flow received at the end of the period t

For bonds that pay interest semiannually, the duration equation becomes:

Macaulay duration

$$D = \frac{\sum_{t=1/2}^N \frac{CF_t \times t}{(1+r/2)^{2t}}}{\sum_{t=1/2}^N \frac{CF_t}{(1+r/2)^{2t}}}$$

where $t = 1/2, 1, 1\frac{1}{2}, \dots, N$.

Example

Suppose that you have a bond that offers a coupon rate of 10 percent paid semiannually (or 5 percent paid every 6 months). The face value of the bond is \$1,000, it matures in four years, its current rate of return (*rb*) is 8 percent, and its current price is \$1,067.34.

Duration of this bond is ????

Duration of a Four-Year Bond with 10 Percent Coupon Paid Semiannually and 8 Percent Yield

t	CF_t	$\frac{1}{(1 + 4\%)^{2t}}$	$\frac{CF_t}{(1 + 4\%)^{2t}}$	$\frac{CF_t \times t}{(1 + 4\%)^{2t}}$
½	50	0.9615	48.08	24.04
1	50	0.9246	46.23	46.23
1½	50	0.8890	44.45	66.67
2	50	0.8548	42.74	85.48
2½	50	0.8219	41.10	102.75
3	50	0.7903	39.52	118.56
3½	50	0.7599	38.00	133.00
4	1,050	0.7307	767.22	3,068.88
			1,067.34	3,645.61

$$D = \frac{3,645.61}{1,067.34} = 3.42 \text{ years}$$

In other words, on a time value of money basis, the initial investment of \$1,067.34 is recovered after 3.42 years.

Class Problem

- Compute duration of a bond with $FV = \$1000$, Coupon rate = 12% paid **semiannually**, interest rate = 10% and maturity period 5 years.
- Compute duration of a bond with $FV = \$1000$, Coupon rate = 10% paid **semiannually**, interest rate = 10% and maturity period 5 years.
- Compute duration of a bond with $FV = \$1000$, Coupon rate = 8% paid **semiannually**, interest rate = 10% and maturity period 5 years.

Class Problem: Solution

- Compute duration of a bond with FV=\$1000, Coupon rate=12% paid semiannually, interest rate= 10% and maturity period 5 years.
- Compute duration of a bond with FV=\$1000, Coupon rate=10% paid semiannually, interest rate= 10% and maturity period 5 years.
- Compute duration of a bond with FV=\$1000, Coupon rate=8% paid semiannually, interest rate= 10% and maturity period 5 years.

Coupon	Interest	Duration
12	10	3.95
10	10	4.05
8	10	4.18

Solution

- Compute duration of a bond with FV=\$1000, Coupon rate=12% paid **semiannually**, interest rate= 10% and maturity period **3 years**.
- Compute duration of a bond with FV=\$1000, Coupon rate=10% paid **semiannually**, interest rate= 10% and maturity period **3 years**.
- Compute duration of a bond with FV=\$1000, Coupon rate=8% paid **semiannually**, interest rate= 10% and maturity period **3 years**.

Coupon	Interest	Duration
12	10	2.61
10	10	2.66
8	10	2.72

Solution

- Compute duration of a bond with FV=\$1000, Coupon rate=12% paid quarterly, interest rate= 10% and maturity period 3 years.
- Compute duration of a bond with FV=\$1000, Coupon rate=10% paid quarterly, interest rate= 10% and maturity period 3 years.
- Compute duration of a bond with FV=\$1000, Coupon rate=8% paid quarterly, interest rate= 10% and maturity period 3 years.

Coupon	Interest	Duration
12	10	2.57
10	10	2.62
8	10	2.36

Summary

		Quarterly	Semiannual
Coupon	Interest	Duration	
12	10	2.57	2.61
10	10	2.62	2.66
8	10	2.36	2.72

The Duration of a Zero-Coupon Bond

- Suppose that you have a zero-coupon bond with a face value of \$1,000, a maturity of four years, and a current rate of return of 8 percent compounded semiannually.

t	CF_t	$\frac{1}{(1 + 8\%/2)^{2 \times 4}}$	$\frac{CF_4}{(1 + 8\%/2)^{2 \times 4}}$	$\frac{CF_t \times 4}{(1 + 8\%/2)^{2 \times 4}}$
4	\$1,000	0.7307	730	2,923

$$D = 2,923/730 = 4 \text{ years}$$

Duration of bond

- Duration is the weighted average of the times to each of the cash payments.
- A measure of the average life that could be used to predict the exposure of each bond's price to fluctuations in interest rates.
- Also known as **Macaulay duration**.

$$\text{Duration} = \frac{1 \times \text{PV}(C_1)}{\text{PV}} + \frac{2 \times \text{PV}(C_2)}{\text{PV}} + \frac{3 \times \text{PV}(C_3)}{\text{PV}} + \dots + \frac{T \times \text{PV}(C_T)}{\text{PV}}$$

$$\text{Modified duration} = \text{volatility}(\%) = \frac{\text{duration}}{1 + \text{yield}}$$

Example

- Compute duration for the 9% seven-year bonds (FV= \$1000) , assuming annual payments. The yield to maturity is 4% a year.

	YTM	4%		
year	Payment	PV of Cf	Fraction of total value	Fraction x Year
1	90	86.53846	0.066562787	0.066562787
2	90	83.21006	0.06400268	0.12800536
3	90	80.00967	0.061541038	0.184623115
4	90	76.93238	0.059174075	0.236696302
5	90	73.97344	0.056898149	0.284490747
6	90	71.12831	0.054709759	0.328258554
7	1090	828.3104	0.637111511	4.459780574
		1300.103		5.69

Other expression for duration

$$D = \frac{\sum_{t=1}^N \frac{CF_t \times t}{(1+r)^t}}{\sum_{t=1}^N \frac{CF_t}{(1+r)^t}} = \frac{\sum_{t=1}^N PV_t \times t}{\sum_{t=1}^N PV_t}$$

where

D = Duration measured in years

t = 1 to T , the period in which a cash flow is received

N = Number of periods to maturity

CF_t = Cash flow received on the security at end of period t

r = Current required rate of return (r) or yield to maturity (ym) on the investment

PV_t = Present value of the cash flow received at the end of the period t

For bonds that pay interest semiannually, the duration equation becomes:

$$D = \frac{\sum_{t=1/2}^N \frac{CF_t \times t}{(1+r/2)^{2t}}}{\sum_{t=1/2}^N \frac{CF_t}{(1+r/2)^{2t}}}$$

where $t = 1/2, 1, 1\frac{1}{2}, \dots, N$.

Example

- Suppose that you have a bond that offers a coupon rate of 10 percent paid semiannually (or 5 percent paid every 6 months). The face value of the bond is \$1,000, it matures in four years, its current rate of return (r_b) is 8 percent, and its current price is \$1,067.34.
- Duration of this bond is ????

Duration of a Four-Year Bond with 10 Percent Coupon Paid Semiannually and 8 Percent Yield

t	CF_t	$\frac{1}{(1 + 4\%)^{2t}}$	$\frac{CF_t}{(1 + 4\%)^{2t}}$	$\frac{CF_t \times t}{(1 + 4\%)^{2t}}$
½	50	0.9615	48.08	24.04
1	50	0.9246	46.23	46.23
1½	50	0.8890	44.45	66.67
2	50	0.8548	42.74	85.48
2½	50	0.8219	41.10	102.75
3	50	0.7903	39.52	118.56
3½	50	0.7599	38.00	133.00
4	1,050	0.7307	767.22	3,068.88
			1,067.34	3,645.61

$$D = \frac{3,645.61}{1,067.34} = 3.42 \text{ years}$$

Previous example

Compute duration for the 9% seven-year bonds (FV= \$1000) , assuming **semi-annual** payments. The yield to maturity is 4% a year.

	Annual	4%		
year	Payment	PV of Cf	Fraction of total value	Fraction x Year
0.5	45	44.1176	0.0339	0.0169
1	45	43.2526	0.0332	0.0332
1.5	45	42.4045	0.0326	0.0488
2	45	41.5730	0.0319	0.0638
2.5	45	40.7579	0.0313	0.0782
3	45	39.9587	0.0307	0.0920
3.5	45	39.1752	0.0301	0.1053
4	45	38.4071	0.0295	0.1179
4.5	45	37.6540	0.0289	0.1301
5	45	36.9157	0.0283	0.1417
5.5	45	36.1918	0.0278	0.1528
6	45	35.4822	0.0272	0.1634
6.5	45	34.7865	0.0267	0.1736
7	1045	791.9794	0.6080	4.2558
	PV	1302.656	Duration	5.573622489

Why duration is important concept?

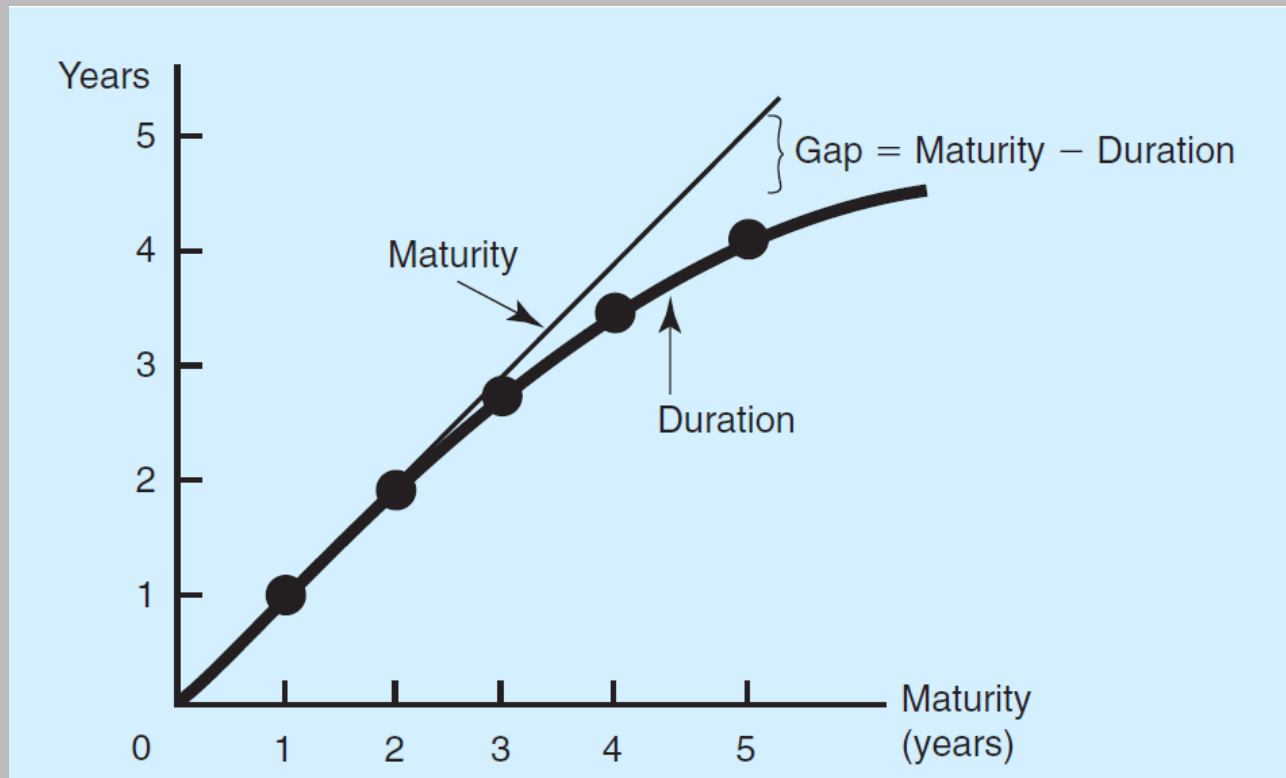
- The effect of maturity and coupon rates on the sensitivity of bond prices to changes in interest rates is complex to deal and understand
- Duration provides a simple measure that allows for a straightforward calculation of a bond's interest rate sensitivity.
- Duration is the weighted - average time to maturity on a financial security using the relative present values of the cash flows as weights. On a time value of money basis, duration measures the weighted average of when cash flows are received on a security.

Some fact about duration

- Compute duration of a bond with FV=\$1000, Coupon rate=12% paid semiannual and quarterly frequency, interest rate=12, 10 and 8% and maturity period 3 years.

		Quarterly	Semiannual
Coupon	Interest	Duration	
12	10	2.57	2.61
10	10	2.62	2.66
8	10	2.36	2.72

Maturity and Duration



The specific relationship between these factors for securities with annual compounding of interest is represented as

$$\frac{\Delta P/P}{\Delta r_b / (1 + r_b)} = -D$$

THE FISHER EFFECT

- The relationship between nominal returns, real returns, and inflation. Let R stand for the nominal rate and r stand for the real rate.

$$1 + R = (1 + r) \times (1 + h)$$

where h is the inflation rate.

$$1 + R = (1 + r) \times (1 + h)$$

$$R = r + h + r \times h$$

This third component is usually small, so it is often dropped. The nominal rate is then approximately equal to the real rate plus the inflation rate:

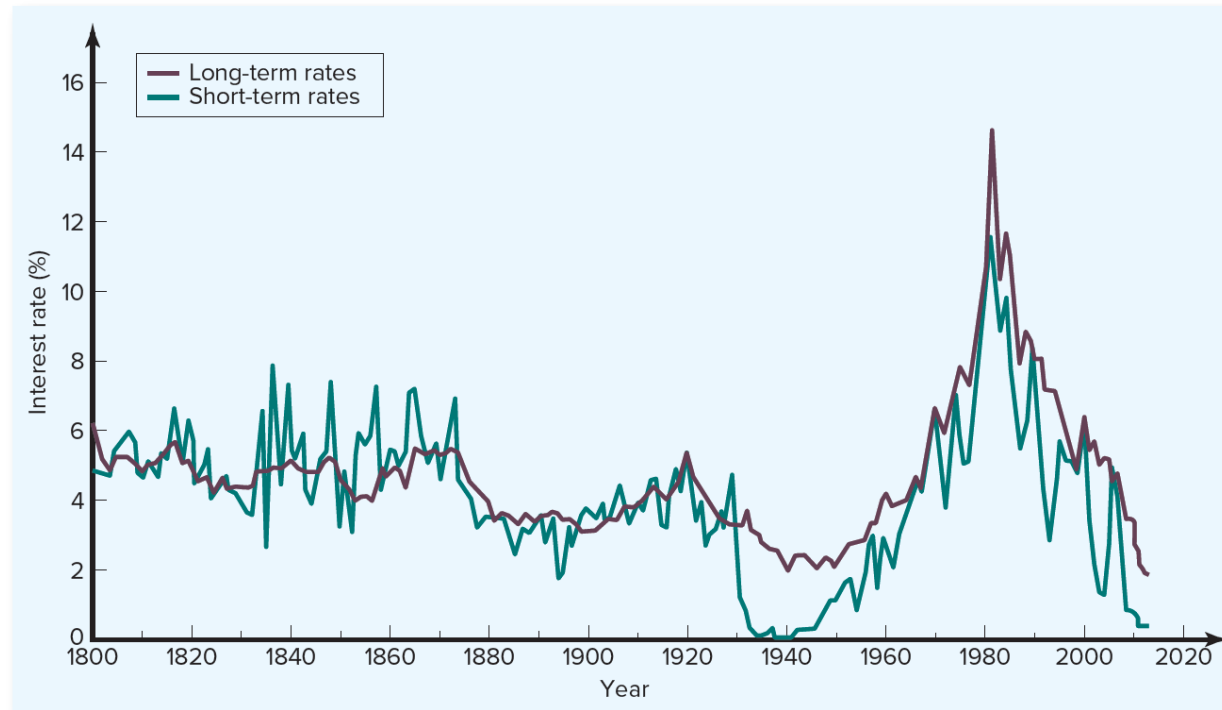
$$R \approx r + h$$

Determinants of Bond Yields

- Term structure of interest rates: The relationship between nominal interest rates on default-free, pure discount securities and time to maturity; that is, the pure time value of money.
- the term structure of interest rates tells us what nominal interest rates are on default-free, pure discount bonds of all maturities.
- **inflation premium** : The portion of a nominal interest rate that represents compensation for expected future inflation.
- **interest rate risk premium** : The compensation investors demand for bearing interest rate risk.

Term Rate

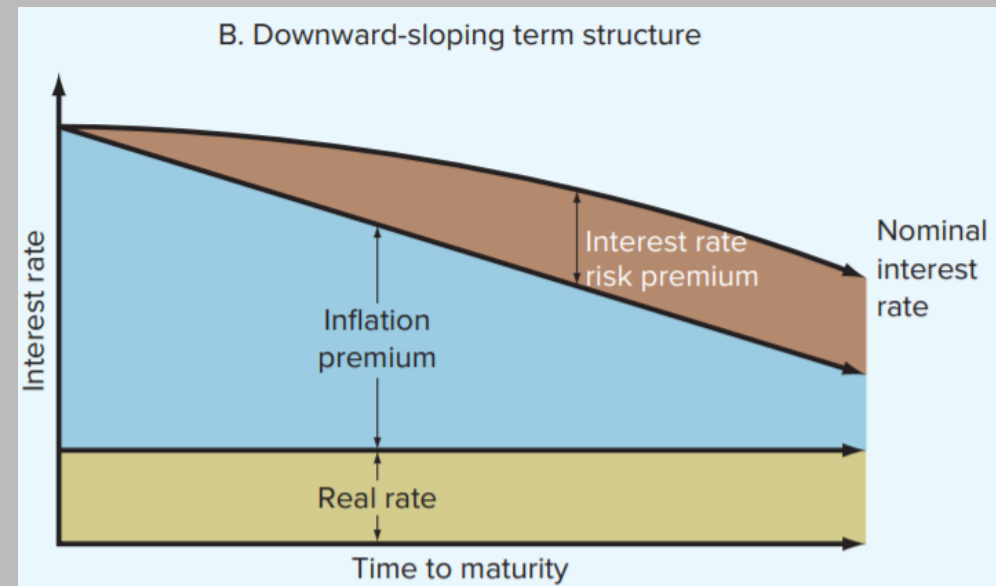
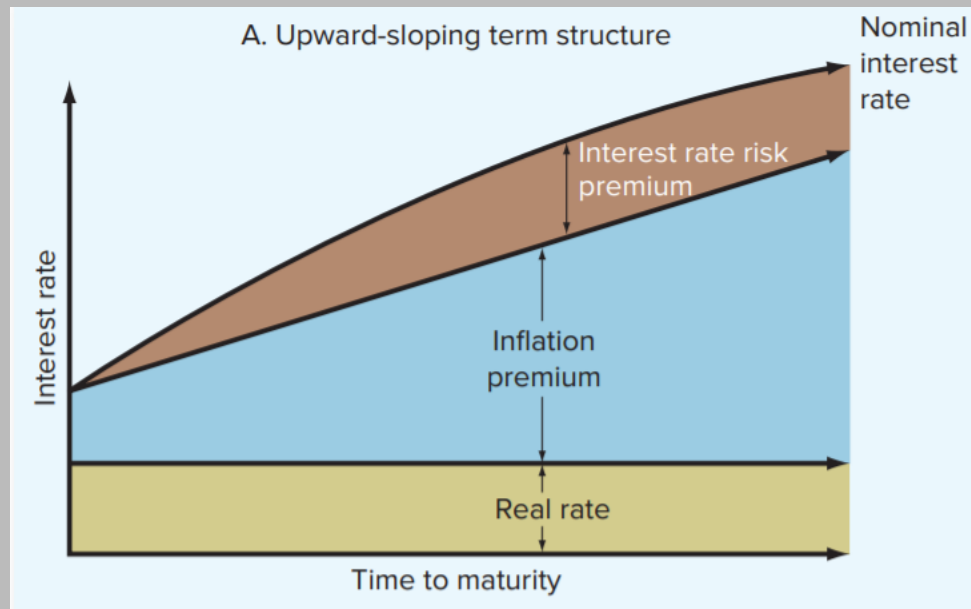
FIGURE 7.5 U.S. Interest Rates: 1800–2016



SOURCE: Siegel, Jeremy J., *Stocks for the Long Run*, 4th edition. New York: McGraw-Hill, 2008, updated by the authors.

Term structure of interest rates

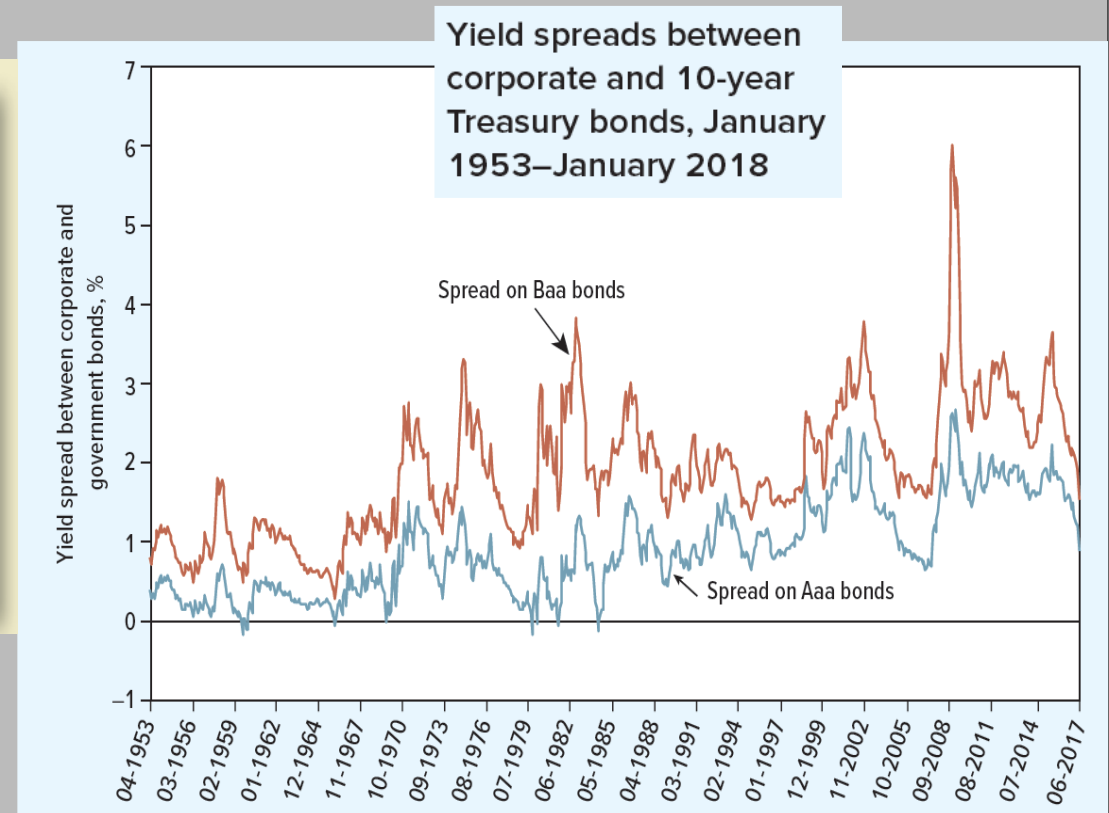
- The relationship between nominal interest rates on default-free, pure discount securities and time to maturity; that is, the pure time value of money.



Corporate and Default Risk

Issuer Name	Coupon (%)	Maturity	S&P Rating	Price (%)	Yield (%)
Johnson & Johnson	6.73	2023	AAA	125.05	2.30
Walmart	6.75	2023	AA	122.95	2.62
Alabama Power	3.55	2023	A	104.74	2.71
PSEG	4.3	2023	BBB	105.65	3.23
Freeport-McMoRan	3.875	2023	BB	117.44	4.41
NGL Energy	7.5	2023	B	99.50	7.60
Quorum Health	11.625	2023	CCC	94.50	13.05

Because of the risk of default, yields on corporate bonds are higher than those of government bonds.



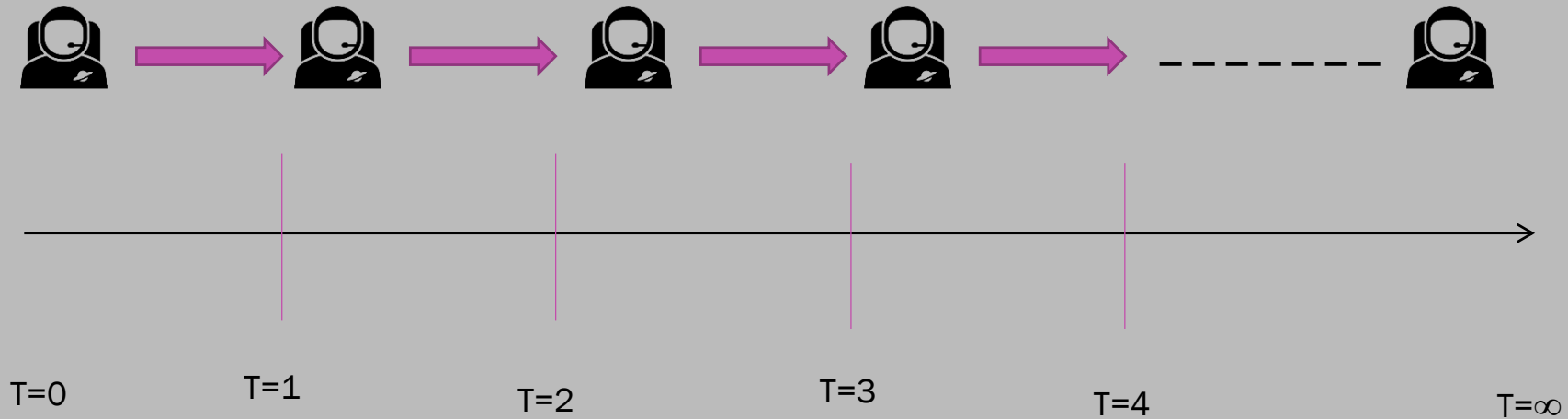
Yield on Bond

SELECTED INTERNATIONAL GOVERNMENT 10-YEAR BOND YIELDS	
	Yield (%)
United States	2.41
United Kingdom	1.19
Japan	.04
Canada	2.04
Australia	2.63
Greece	4.03
Spain	1.55
Italy	2.00
India	7.32

SOURCE: Data pulled from www.bloomberg.com, December 30, 2017.

Stock Valuation

Value of a stock over long time horizon



Cash Flows for Stockholders

- If you buy a share of stock, you can receive cash in two ways:
 - The company pays dividends.
 - You sell your shares, either to another investor in the market or back to the company.
- As with bonds, the price of the stock is the present value of these expected cash flows.

Example: Cash flow from Stock

If you buy the stock today and sell it at the end of the year, you will have a total of \$80 in cash. At 25 percent:

$$\text{Present value} = (\$10 + 70)/1.25 = \$64$$

Therefore, \$64 is the value you would assign to the stock today.

$$P_0 = (D_1 + P_1)/(1 + R)$$

where R is the required return in the market on this investment.

$$P_1 = (D_2 + P_2)/(1 + R)$$

$$\begin{aligned} P_0 &= \frac{D_1 + P_1}{1 + R} = \frac{D_1 + \frac{D_2 + P_2}{1 + R}}{1 + R} \\ &= \frac{D_1}{(1 + R)^1} + \frac{D_2}{(1 + R)^2} + \frac{P_2}{(1 + R)^2} \end{aligned}$$

One-Period Example

- Suppose you are thinking of purchasing the stock of Moore Oil, Inc.
 - You expect it to pay a \$2 dividend in one year, and you believe that you can sell the stock for \$14 at that time.
 - If you require a return of 20% on investments of this risk, what is the maximum you would be willing to pay?
- Compute the PV of the expected cash flows.
- $\text{Price} = (14 + 2) / (1.2) = \13.33

Two-Period Example

- Now, what if you decide to hold the stock for two years?
 - In addition to the dividend in one year, you expect a dividend of \$2.10 in two years and a stock price of \$14.70 at the end of year 2.
 - Now how much would you be willing to pay?
- $PV = 2 / (1.2) + (2.10 + 14.70) / (1.2)^2 = 13.33$

Three-Period Example

- Finally, what if you decide to hold the stock for three years?
 - In addition to the dividends at the end of years 1 and 2, you expect to receive a dividend of \$2.205 at the end of year 3 and the stock price is expected to be \$15.435.
 - Now how much would you be willing to pay?
 - $PV = 2 / 1.2 + 2.10 / (1.2)^2 + (2.205 + 15.435) / (1.2)^3 = 13.33$

Generalised form

- Three period-

$$\begin{aligned}P_0 &= \frac{D_1}{(1+R)^1} + \frac{D_2}{(1+R)^2} + \frac{P_2}{(1+R)^2} \\&= \frac{D_1}{(1+R)^1} + \frac{D_2}{(1+R)^2} + \frac{\frac{D_3 + P_3}{1+R}}{(1+R)^2} \\&= \frac{D_1}{(1+R)^1} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{P_3}{(1+R)^3}\end{aligned}$$

$$P_0 = \frac{D_1}{(1+R)^1} + \frac{D_2}{(1+R)^2} + \frac{D_3}{(1+R)^3} + \frac{D_4}{(1+R)^4} + \frac{D_5}{(1+R)^5} + \dots$$

Estimating Dividends: Special Cases

- Constant dividend (i.e., zero growth)
 - The firm will pay a constant dividend forever.
 - This is like preferred stock.
 - The price is computed using the perpetuity formula.
- Constant dividend growth
 - The firm will increase the dividend by a constant *percent* every period.
 - The price is computed using the growing perpetuity model.
- Supernormal growth
 - Dividend growth is not consistent initially, but settles down to constant growth eventually.
 - The price is computed using a multistage model.

Zero Growth

- If dividends are expected at regular intervals forever, then this is a perpetuity, and the present value of expected future dividends can be found using the **perpetuity** formula.
 - $P_0 = D / R$
- Suppose a stock is expected to pay a \$0.50 dividend every quarter and the required return is 10% with quarterly compounding. What is the price?
 - $P_0 = .50 / (0.1 / 4) = \20

$$D_1 = D_2 = D_3 = D = \text{constant}$$

So, the value of the stock is:

$$P_0 = \frac{D}{(1+R)^1} + \frac{D}{(1+R)^2} + \frac{D}{(1+R)^3} + \frac{D}{(1+R)^4} + \frac{D}{(1+R)^5} + \dots$$

Because the dividend is always the same, the stock can be viewed as an ordinary perpetuity with a cash flow equal to D every period. The per-share value is given by:

$$P_0 = D/R$$

where R is the required return.

Dividend Growth Model

- Dividends are expected to grow at a constant percent per period.

- $P_0 = D_1 / (1+R) + D_2 / (1+R)^2 + D_3 / (1+R)^3 + \dots$

- $P_0 = D_0(1+g)/(1+R) + D_0(1+g)^2/(1+R)^2 + D_0(1+g)^3/(1+R)^3 + \dots$

$$P_0 = \frac{D_0 \times (1+g)}{R-g} = \frac{D_1}{R-g}$$

- this reduces to:

$$P_0 = \frac{D_0(1+g)}{R-g} = \frac{D_1}{R-g}$$

$$D_1 = D_0 \times (1+g)$$

The dividend in two periods is:

$$\begin{aligned} D_2 &= D_1 \times (1+g) \\ &= [D_0 \times (1+g)] \times (1+g) \\ &= D_0 \times (1+g)^2 \end{aligned}$$

Example

- The next dividend for the Gordon Growth Company will be \$4 per share. Investors require a 16 percent return on companies such as Gordon. Gordon's dividend increases by 6 percent every year. Based on the dividend growth model, what is the value of Gordon's stock today? What is the value in four years?

Solution

$$\begin{aligned}P_0 &= D_1 / (R - g) \\&= \$4 / (.16 - .06) \\&= \$4 / .10 \\&= \$40\end{aligned}$$

Because we already have the dividend in one year, we know that the dividend in four years is equal to $D_1 \times (1 + g)^3 = \$4 \times 1.06^3 = \4.764 . So, the price in four years is:

$$\begin{aligned}P_4 &= D_4 \times (1 + g) / (R - g) \\&= \$4.764 \times 1.06 / (.16 - .06) \\&= \$5.05 / .10 \\&= \$50.50\end{aligned}$$

Notice in this example that P_4 is equal to $P_0 \times (1 + g)^4$.

$$P_4 = \$50.50 = \$40 \times 1.06^4 = P_0 \times (1 + g)^4$$

To see why this is so, notice first that:

$$P_4 = D_5 / (R - g)$$

However, D_5 is just equal to $D_1 \times (1 + g)^4$, so we can write P_4 as:

$$\begin{aligned}P_4 &= D_1 \times (1 + g)^4 / (R - g) \\&= [D_1 / (R - g)] \times (1 + g)^4 \\&= P_0 \times (1 + g)^4\end{aligned}$$

Nonconstant Growth

Example - I

- Suppose a firm is expected to increase dividends by 20% in one year and by 15% in two years.
- After that, dividends will increase at a rate of 5% per year indefinitely.
- If the last dividend was \$1 and the required return is 20%, what is the price of the stock?
- Remember that we have to find the PV of all expected future dividends.

Nonconstant Growth Example - II

- Compute the dividends until growth levels off.
 - $D_1 = 1(1.2) = \$1.20$
 - $D_2 = 1.20(1.15) = \$1.38$
 - $D_3 = 1.38(1.05) = \$1.449$
- Find the expected future price.
 - $P_2 = D_3 / (R - g) = 1.449 / (.2 - .05) = 9.66$
- Find the present value of the expected future cash flows.
 - $P_0 = 1.20 / (1.2) + (1.38 + 9.66) / (1.2)^2 = 8.67$

Using the DGM to Find R

- Start with the DGM:

$$P_0 = \frac{D_0(1+g)}{R-g} = \frac{D_1}{R-g}$$

$$R = \frac{D_0(1+g)}{P_0} + g = \frac{D_1}{P_0} + g$$

Stock Valuation Using Multiples

- Another common valuation approach is to multiply a benchmark PE ratio by earnings per share (EPS) to come up with a stock price.
- $P_t = \text{Benchmark PE ratio} \times \text{EPS}_t$
- The benchmark PE ratio is often an industry average or based on a company's own historical values.
- The price-sales ratio can also be used.

Exercise- E1

- A bond has a quoted price of \$1,080.42. It has a face value of \$1,000, a semiannual coupon of \$30, and a maturity of five years. What is its current yield? What is its yield to maturity? Which is bigger? Why?

Exercise- E1

- A bond has a quoted price of \$1,080.42. It has a face value of \$1,000, a semiannual coupon of \$30, and a maturity of five years. What is its current yield? What is its yield to maturity? Which is bigger? Why?
- Solution:-
- The current yield is thus $\$60/\$1,080.42 = 5.55 \text{ percent}$
- the yield to maturity. In this case, the bond pays \$30 every six months and has 10 six-month periods until maturity.

$$\$1,080.42 = \$30 \times [1 - 1/(1 + r)^{10}]/r + 1,000/(1 + r)^{10}$$

r is equal to about 2.1 percent. This 2.1 percent is the yield per six months. yield to maturity is 4.2 percent (2x2.1%), which is less than the current yield.

Exercise-2

- Suppose taxable bonds are currently yielding 8 percent, while at the same time, munis of comparable risk and maturity are yielding 6 percent. Which is more attractive to an investor in a 40 percent bracket? What is the break-even tax rate? How do you interpret this rate?

Solution

- For an investor in a 40 percent tax bracket, a taxable bond yields $8 \times (1 - .40) = 4.8$ percent after taxes, so the muni is much more attractive. The break-even tax rate is the tax rate at which an investor would be indifferent between a taxable and a nontaxable issue. If we let t^* stand for the break-even tax rate, then we can solve for it as follows:

$$.08 \times (1 - t^*) = .06$$

$$1 - t^* = .06/.08 = .75$$

$$t^* = .25$$

An investor in a 25 percent tax bracket would make 6 percent after taxes from either bond.

Problem for students 1

The Premier Instruments Ltd (PIL) had paid the following dividends per share.

<i>Year</i>	<i>Dividend per share</i>	<i>Year</i>	<i>Dividend per share</i>
6	Rs 2.80	3	2.24
5	2.58	2	2.10
4	2.40	1	2.00

Assuming a 16 per cent required return, and Rs 3 per share dividend in year 7 (D_1) compute the value of the shares of PIL.

Solution

Year	Dividend	Value of Dividend 7th years onwards	PV of Dividened
1	2		1.724137931
2	2.1		1.560642093
3	2.24		1.435073189
4	2.4		1.325498635
5	2.58		1.22837158
6	2.8	33.86004515	1.149238313
	33.86005		13.89759327
			22.32055501

Problem : YTM

- The market price of a Rs.1,000 par value bond carrying a coupon rate of 14 percent and maturing after 5 years in Rs.1050. What is the yield to maturity (YTM) on this bond? What is the approximate YTM?

Solution

The YTM is the value of r in the following equation:

$$\begin{aligned} 1,050 &= \sum_{t=1}^5 \frac{140}{(1+r)^t} + \frac{1,000}{(1+r)^5} \\ &= 140 (\text{PVIFA}_{r, 5 \text{ yrs}}) + 1,000 (\text{PVIF}_{r, 5 \text{ yrs}}) \end{aligned}$$

$$\begin{aligned} &140 (\text{PVIFA}_{13\%, 5 \text{ yrs}}) + 1,000 (\text{PVIF}_{13\%, 5 \text{ yrs}}) \\ &= 140 (3.517) + 1,000 (0.543) \\ &= 492.4 + 543.0 = \text{Rs.}1035.4 \end{aligned}$$

$$\begin{aligned} &140 (\text{PVIFA}_{12\%, 5 \text{ yrs}}) + 1,000 (\text{PVIF}_{12\%, 5 \text{ yrs}}) \\ &= 140 (3.605) + 1,000 (0.567) \\ &= 504.7 + 567.0 = \text{Rs.}1071.7 \end{aligned}$$

$$12\% + (13\% - 12\%) \frac{1071.7 - 1050.0}{1071.7 - 1035.4} = 12.60 \text{ percent}$$

(b) The approximate YTM works out to:

$$\text{YTM} = \frac{140 + (1,000 - 1,050)/5}{0.40 \times 1000 + 0.6 \times 1050} = 12.62 \text{ percent}$$

Problem :Stock

- The equity stock of Rax Limited is currently selling for Rs.30 per share. The dividend expected next year is Rs.2.00. The investors' required rate of return on this stock is 15 percent. If the constant growth model applies to Rax Limited, what is the expected growth rate ?

Solution

According to the constant growth model

$$P_0 = \frac{D_1}{r - g}$$

This means

$$g = r - D_1 / P_0$$

Hence, the expected growth rate (g) for Rax Limited is:

$$g = 0.15 - \frac{2.00}{30.00} = .083 \text{ or } 8.3 \text{ percent}$$

Problem : Stock

- Vardhman Limited's earnings and dividends have been growing at a rate of 18 percent per annum. This growth rate is expected to continue for 4 years. After that the growth rate will fall to 12 percent for the next 4 years. Thereafter, the growth rate is expected to be 6 percent forever. If the last dividend per share was Rs.2.00 and the investors' required rate of return on Vardhman's equity is 15 percent, what is the intrinsic value per share?

Solution

Step 1: The dividend stream during the first eight years when Vardhman would enjoy a relatively high rate of growth will be:

$$D_1 = 2.00 (1.18) = 2.36$$

$$D_2 = 2.00 (1.18)^2 = 2.78$$

$$D_3 = 2.00 (1.18)^3 = 3.29$$

$$D_4 = 2.00 (1.18)^4 = 3.88$$

$$D_5 = 2.00 (1.18)^4 (1.12) = 4.34$$

$$D_6 = 2.00 (1.18)^4 (1.12)^2 = 4.86$$

$$D_7 = 2.00 (1.18)^4 (1.12)^3 = 5.45$$

$$D_8 = 2.00 (1.18)^4 (1.12)^4 = 6.10$$

$$\begin{aligned} & 2.36 (\text{PVIF}_{15\%, 1 \text{ yrs}}) + 2.78 (\text{PVIF}_{15\%, 2 \text{ yrs}}) \\ & + 3.29 (\text{PVIF}_{15\%, 3 \text{ yrs}}) + 3.88 (\text{PVIF}_{15\%, 4 \text{ yrs}}) \\ & + 4.34 (\text{PVIF}_{15\%, 5 \text{ yrs}}) + 4.86 (\text{PVIF}_{15\%, 6 \text{ yrs}}) \\ & + 5.45 (\text{PVIF}_{15\%, 7 \text{ yrs}}) + 6.10 (\text{PVIF}_{15\%, 8 \text{ yrs}}) \\ & = 2.36 (0.870) + 2.78 (0.756) + 3.29 (0.658) \\ & + 3.88 (0.572) + 4.34 (0.497) + 4.86 (0.432) \\ & + 5.45 (0.376) + 6.10 (0.327) \\ & = 2.05 + 2.10 + 2.16 + 2.22 + 2.16 + 2.10 + 2.05 + 1.99 = \text{Rs.16.83} \end{aligned}$$

Step 2: The price of the share at the end of 8 years, applying the constant growth model at that point of time, will be:

$$\begin{aligned} P_8 &= \frac{D_9}{r - g_n} = \frac{D_8(1 + g_n)}{r - g_n} \\ &= \frac{2.00(1.18)^4(1.12)^4(1.06)}{0.15 - 0.06} = \text{Rs.}71.84 \end{aligned}$$

The present value of this price is:

$$\frac{71.84}{(1.15)^8} = 23.49$$

Step 3 : The sum of the above components is:

$$P_0 = \text{Rs.}16.83 + \text{Rs.}23.49 = \text{Rs.}40.32$$



END OF SESSION

Stock valuation